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Project Report

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EcoSort - AI

AI-Based Waste Classification and Recycling Assistant Using MobileNetV2

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1. Introduction:

EcoSort AI is an AI-based waste classification and recycling assistant developed using deep learning. The purpose of this project is to identify different types of waste from images and help users understand how waste can be sorted for recycling.

The system uses the MobileNetV2 deep learning model to classify waste images into six categories: cardboard, glass, metal, paper, plastic and trash. A user-friendly web interface allows users to upload an image and view the model's predicted category along with confidence scores and recycling recommendations.

This project aims to demonstrate how artificial intelligence can support waste management by making waste classification easier, faster and more accessible. EcoSort AI can contribute to environmental awareness and encourage better waste-sorting practices.

2. Problem Statement:

Waste management is a major environmental challenge because different types of waste require different methods of disposal and recycling. Incorrect waste sorting can make recycling more difficult and increase the amount of waste sent to landfills. Manually identifying and separating waste can also be time-consuming and requires basic knowledge of different waste categories.

EcoSort AI addresses this problem by using deep learning to classify waste images into six categories: cardboard, glass, metal, paper, plastic and trash. The system provides predicted categories, confidence scores and recycling recommendations through a user-friendly interface.

The main problem addressed by this project is the need for a simple and accessible AI-based system that assists users in identifying waste and supports more effective waste-sorting practices.

3. Research Objectives:

The main objective of EcoSort AI is to develop an AI-based system that can automatically classify waste images into different categories and provide useful recycling recommendations.
The specific objectives of this project are:

1. To develop a deep learning model using MobileNetV2 for waste image classification.
2. To classify waste into six categories: cardboard, glass, metal, paper, plastic and trash.
3. To preprocess and prepare the dataset for effective model training.
4. To evaluate the model's performance using validation accuracy and loss.
5. To develop a user-friendly web interface using Gradio for uploading images and displaying classification results.
6. To provide category-specific recycling recommendations to encourage proper waste management.
7. To explore the use of artificial intelligence in improving waste identification and environmental awareness.

4. Literature Review:

Artificial intelligence and deep learning have become important technologies for image classification and automated waste management. Traditional waste sorting often depends on manual identification which can be time-consuming and may lead to incorrect classification. Computer vision techniques provide an alternative by allowing computers to recognize and classify objects from images.

Convolutional Neural Networks (CNNs) are widely used for image classification because they can learn visual features such as shapes, edges, textures and patterns. However, training a deep learning model from scratch can require substantial computational resources and a large amount of training data.

MobileNetV2 is a lightweight convolutional neural network designed to provide efficient image recognition while reducing computational requirements. It’s pretrained version can be used through transfer learning, allowing a model to benefit from features learned from a large image dataset and adapt them to a specific task.

In EcoSort AI, MobileNetV2 is used to classify waste images into six categories: cardboard, glass, metal, paper, plastic and trash. The model is integrated into a Gradio web interface that displays predictions and confidence scores and provides recycling recommendations. This project demonstrates a practical application of deep learning for waste identification and environmental awareness.

5. Tools and Technologies Used:

The development of EcoSort AI involved several tools, programming libraries and technologies for dataset preparation, deep learning model development, training, evaluation and user interface implementation.

1. Python: Python was used as the primary programming language for developing the waste classification system.
2. TensorFlow and Keras: These libraries were used to build, train, save, load and evaluate the deep learning model.
3. MobileNetV2: A pretrained convolutional neural network was used through transfer learning to classify waste images into six categories.
4. NumPy: NumPy was used to handle image arrays and prepare numerical data for model predictions.
5. Pillow (PIL): Pillow was used to open, resize, and convert uploaded images into the format required by the model.
6. Gradio: Gradio was used to develop the web interface allowing users to upload waste images and view classification predictions and recycling recommendations.
7. Jupyter Notebook: Jupyter Notebook was used to write and execute Python code, prepare the dataset, train the model and evaluate its performance.
8. Pretrained ImageNet Weights: ImageNet pretrained weights were used to initialize MobileNetV2 with previously learned visual features, supporting transfer learning for waste classification.

6. Dataset Description:

The EcoSort AI project uses a garbage classification dataset containing images of different types of waste. The dataset was used to train and evaluate the deep learning model for automatic waste image classification.

The dataset contains a total of 2,527 images divided into six categories: cardboard, glass, metal, paper, plastic and trash. Each category represents a different type of waste that the model learns to identify based on visual features such as shape, texture, and appearance.
The distribution of images across the six categories is shown below:

Waste Category	**Number of Images**
Cardboard	403
Glass	501
Metal	410
Paper	594
Plastic	482
Trash	137
Total	**2,527**

The dataset was organized into category-specific folders so that the images could be loaded and labeled automatically during model training. It was divided into training and validation subsets using an 80:20 split. Approximately 80% of the images were used for training while the remaining 20% were used for validation.
This dataset enabled the model to learn patterns associated with different waste categories and provided a basis for evaluating its classification performance.

7. Methodology:

The methodology of EcoSort AI describes the steps followed to develop an AI-based waste classification system. The project uses transfer learning with MobileNetV2 to classify waste images into six categories.
The development process consists of the following steps:
1. Dataset Collection: A garbage classification dataset containing 2,527 images across six waste categories was selected for the project.
2. Dataset Preparation: The images were organized into category-specific folders and divided into training and validation subsets using an 80:20 split.
3. Image Preprocessing: The images were resized to 160 × 160 pixels, converted into RGB format where required, and preprocessed according to MobileNetV2's input requirements.
4. Model Development: A pretrained MobileNetV2 model was used as the base feature extractor. Additional layers were added to classify the images into the six waste categories.
5. Model Training: The classification layers were trained using the prepared training dataset. Early stopping and learning-rate adjustment techniques were used to help improve training.
6. Model Evaluation: The trained model was evaluated on the validation subset using accuracy and loss to measure its classification performance.
7. Interface Development: A web interface was developed using Gradio so users could upload images and receive predicted categories, confidence scores, and recycling recommendations.
8. Final Prediction: When a user uploads an image, the system preprocesses it, passes it to the trained model, and displays the predicted waste category and corresponding recycling advice.

8. Data Preprocessing:

Data preprocessing is an important step in EcoSort AI because the images must be prepared in a consistent format before they can be processed by the deep learning model.
The following preprocessing steps were used in the project:

1. Image Resizing: All images were resized to 160 × 160 pixels to ensure a consistent input size for the MobileNetV2 model.
2. RGB Conversion: Images were processed in RGB format, containing three color channels: red, green and blue. This ensured compatibility with the model's input requirements.
3. Image Preprocessing: The MobileNetV2 preprocessing function was applied to transform pixel values into the range expected by the pretrained model.
4. Dataset Splitting: The dataset was divided into training and validation subsets using an 80:20 split. The training subset contained approximately 2,022 images while the validation subset contained approximately 505 images.
5. Automatic Labeling: Images were assigned labels according to their respective folder names. The six labels corresponded to cardboard, glass, metal, paper, plastic and trash.
6. Batch Processing: Images were loaded in batches of 32 to support efficient model training and evaluation.

These preprocessing steps helped standardize the image data and prepare it for training and evaluating the MobileNetV2-based waste classification model.

9. Model Development:

The EcoSort AI system was developed using MobileNetV2, a lightweight convolutional neural network designed for efficient image classification. The project uses transfer learning, which allows a pretrained model to be adapted to a new classification task.

The MobileNetV2 model was initialized with pretrained ImageNet weights, enabling it to use previously learned visual features. Its original classification layer was removed, and the base model was used as a feature extractor by freezing its weights during the initial training phase.

Additional layers were added to perform waste classification. Global Average Pooling was used to reduce the extracted feature maps, followed by a Dropout layer to help reduce overfitting. A fully connected Dense layer with 128 neurons and ReLU activation was added to learn task-specific features. Another Dropout layer was included before the final classification layer.

The output layer contains six neurons with a Softmax activation function, representing the six waste categories: cardboard, glass, metal, paper, plastic, and trash. The Softmax function produces a probability score for each category, and the category with the highest score is selected as the model's prediction.

The model was compiled using the Adam optimizer with a learning rate of 0.001 and sparse categorical cross-entropy as the loss function. Accuracy was used as an evaluation metric.

This architecture combines the feature-extraction capabilities of MobileNetV2 with additional classification layers to provide an efficient solution for waste image classification.

10. Model Training:

The EcoSort AI model was trained using the prepared garbage classification dataset and the MobileNetV2 transfer learning approach. The training process aimed to enable the model to recognize visual differences among six waste categories.

The dataset was divided into training and validation subsets using an 80:20 split. The images were loaded in batches of 32 and resized to 160 × 160 pixels before being passed to the model.

During training, the pretrained MobileNetV2 base was kept frozen, while the additional classification layers learned to distinguish between the waste categories. The model was trained for a maximum of 10 epochs using the Adam optimizer with an initial learning rate of 0.001. Sparse categorical cross-entropy was used as the loss function, and accuracy was monitored throughout training.

Early stopping was implemented to help prevent unnecessary training when improvement stalled, while ReduceLROnPlateau adjusted the learning rate when the validation loss stopped improving. The best model weights were restored according to the early-stopping configuration.

After training, the model was evaluated on the validation subset and saved in Keras format as a .keras file. This saved model was then used in the Gradio interface to classify uploaded waste images.

The training process produced a validation accuracy of approximately 82.97%, providing a useful measure of the model's performance on the held-out validation data.

11. Testing, Validation and Results:

The EcoSort AI model was evaluated to measure its ability to classify waste images into the six target categories. Testing focused on verifying the model's prediction process, evaluating its performance on the validation dataset, and checking its integration with the Gradio web interface.

1. Model Evaluation: The trained MobileNetV2 model was evaluated using the validation subset containing approximately 505 images. The evaluation measured classification accuracy and loss.
2. Evaluation Results: The model achieved a validation accuracy of approximately 82.97% and a validation loss of approximately 0.4616. These results indicate that the model learned useful visual patterns for distinguishing between different waste categories.
3. Image Classification Testing: Images were uploaded through the web interface to check whether the model could generate predictions for cardboard, glass, metal, paper, plastic, and trash. The system displayed predicted categories and confidence scores for the uploaded images.
4. Interface Testing: The Gradio interface was checked to verify that users could upload images, initiate classification, view prediction results and receive category-specific recycling recommendations.
5. Result Analysis: The evaluation indicates that the model can classify waste images with promising performance. However, the accuracy may vary depending on image quality, lighting, background, and similarities between waste categories. Further testing on an independent dataset would be required to assess its generalization to real-world images.

Overall, the results demonstrate that MobileNetV2 can serve as a practical foundation for an AI-based waste classification and recycling assistant.

12. System Implementation and User Interface:

EcoSort AI was implemented as an interactive web application using Python, TensorFlow, Keras, and Gradio. The trained MobileNetV2 model was integrated into the application to allow users to classify waste images through a simple and user-friendly interface.

The interface provides an image upload area where users can select an image of waste from their device. After uploading the image, users can click the classification button to start the prediction process. The application preprocesses the image according to the model's input requirements and passes it to the trained model.

The model generates prediction scores for the six waste categories: cardboard, glass, metal, paper, plastic, and trash. The application displays the classification results with confidence scores to help users understand the model's prediction.

In addition to image classification, the application provides recycling recommendations based on the predicted category. These recommendations offer general guidance on handling different types of waste and encourage responsible waste-sorting practices.

The interface was designed with an emerald-green color theme to reflect the project's focus on environmental protection and sustainability.

The integration of the trained model with Gradio makes EcoSort AI accessible through a web browser without requiring users to interact directly with the underlying Python code.

13. Limitations and Future Enhancements:

Although EcoSort AI provides an effective approach to waste image classification, the current system has certain limitations that can be addressed in future development.

Limitations

1. **Limited Dataset:** The model was trained on a dataset containing 2,527 images across six categories. A larger and more diverse dataset could improve its ability to recognize different waste items.
2. **Class Imbalance:** The number of images varies across categories, with trash having fewer examples than the other categories. This may affect classification performance for underrepresented classes.
3. **Classification Accuracy:** The model achieved approximately 82.97% validation accuracy, meaning that some images may still be classified incorrectly.
4. **Image Quality:** Poor lighting, blurry images, unusual angles, and complex backgrounds may affect prediction accuracy.
5. **Limited Categories:** The current system recognizes only six waste categories and does not identify every possible type of waste.
6. **General Recycling Advice:** The recommendations provide general guidance and may not reflect local recycling facilities or regulations.

Future Enhancements

1. **Dataset Expansion:** Add more diverse images to improve the model's ability to recognize waste under different conditions.
2. **Model Improvement:** Experiment with fine-tuning MobileNetV2 and other suitable deep learning architectures to improve classification performance.
3. **Additional Categories:** Expand the system to recognize more waste types, such as batteries, electronic waste, and organic waste, using appropriately labeled training data.
4. **Mobile Application:** Develop a mobile application to make waste classification accessible on smartphones.
5. **Improved Recycling Recommendations:** Provide more detailed recommendations based on local recycling facilities and waste disposal guidelines.
6. **Real-Time Classification:** Explore camera-based classification to identify waste directly from a live camera feed.

These improvements could make EcoSort AI more accurate, versatile, and useful for supporting responsible waste management.

14. Conclusion:

EcoSort AI is an AI-based waste classification and recycling assistant developed using deep learning and the MobileNetV2 architecture. The project demonstrates how artificial intelligence can be applied to identify different types of waste from images and support better waste-sorting practices.

The system classifies waste images into six categories: cardboard, glass, metal, paper, plastic, and trash. By using transfer learning with pretrained ImageNet weights, the project developed a classification model that achieved approximately 82.97% validation accuracy. The trained model was integrated into a Gradio web interface that allows users to upload waste images, view predicted categories and confidence scores, and receive recycling recommendations.

Although the system has limitations, including dataset imbalance and possible classification errors, it provides a useful foundation for further development. Future improvements could include expanding the dataset, adding more waste categories, fine-tuning the model, and developing a mobile application.

In conclusion, EcoSort AI demonstrates the potential of deep learning to support waste identification, improve environmental awareness, and encourage more responsible recycling practices.

15. References:

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